

No-Code Deep Learning Text Classification for Airline Customer Feedback Using Machine Learning

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Abstract— This research has demonstrated a no-code deep learning approach for determining whether airline passenger comments are positive (Positive_Praise), negative (Negative_Complaint), or urgent (Urgent_Support) based on a classification model trained by a feedforward neural network. Customer comments regarding airlines were categorized into these three areas and then used as input to develop and iterate the classification model for three different sets of experiments. In the first set of experiments, all three categories were assigned an equal amount of training information. The next two sets of experiments added additional training information to determine the impact of the amount of labeled information and the effect of unbalanced datasets on the effectiveness of the model. Model effectiveness was tested with three types of test cases: obvious patterns, sarcasm or ambiguity, and out-of-vocabulary. The data collected during this series of experiments demonstrated that as more training information is provided to the model, the model's ability to classify complex passenger comments improves. Furthermore, when the amounts of training information are distributed equally among the classes of information being classified, the model tends to be less biased toward one classification over another and produces more consistent predictions. As such, this activity provides a practical example of how a neural network learns from labeled examples and highlights the significance of providing high-quality data when developing a reliable text classification model.

Keywords—deep learning, neural networks, text classification, airline customer feedback, machine learning

I. INTRODUCTION

Deep Learning is an area of Artificial Intelligence that allows computers to discover useful relationships within their data without the need for programmers to specify how to do so. Many are using this technique, with one of its most used functions being text classification (i.e., taking a piece of text as input and returning one of several predetermined categories). The use of Deep Learning in text classification has also grown significantly in Customer Service areas due to many companies receiving significant amounts of unorganized textual feedback. Deep Learning models use examples instead of explicit programming to find similarities between languages and thus can aid in quicker decision-making.

The reason we chose the airline sector for our study is due to its continuous generation of consumer feedback on trip experiences, service quality, baggage handling, complaints about bookings, and requests for help. Deep learning-based text classification will enable better identification of customer needs by airlines, improved service quality, and a higher priority placed on critical issues. In previous research,

it has been demonstrated that there are many ways deep learning models can be used to analyze airline passenger reviews, allowing airlines to make decisions based on data analysis (i.e., enhancing consumer satisfaction or informing business-related decision-making) [1]. Therefore, airline customer comments provide us with a suitable source for illustrating how labeled samples can be used to demonstrate the performance of deep learning in text classification.

The goal of this lab project is to show that a no-code deep learning model can be used to categorize airline customer comments into one of three pre-defined categories: Urgent_Support, Negative_Complaint, and Positive_Praise. To determine the effects of the dataset size (and variety) on model results, we performed three different versions of this experiment with progressively larger amounts of training data. We tested the trained model's ability to correctly classify its input in each of these experiments using an unambiguous pattern, an ambiguous or sarcastic comment, and out-of-terminology comments.

II. METHODOLOGY

A. Data Collection and Tagging

The research design utilized a standard process flow diagram with steps of data acquisition and tagging, training models, experimental validation of trained models, assessing performance metrics for validated models, and refinement of datasets (as illustrated in Fig. 1). The research chose airline customer comments as the subject matter due to the variety of comments from customers on airlines (e.g., comments of praise, complaints, or calls for help), which are easily obtainable. The researchers established three different categorizations of classifications: Positive_Praise, Negative_Complaint, and Urgent_Support. Feedback from customers on airlines was hand-tagged by researchers to classify them into one of the categories and to validate that each example demonstrated an accurate representation of the respective classification.

The researchers developed the dataset over three separate rounds of experiments. The first round had 10 tagged examples per classification as a base dataset. Round two added additional examples to the Negative_Complaint and Urgent_Support classifications at a rate of 5 additional examples per classification, leaving Positive_Praise unchanged at 10 examples. As a result of this addition, Round Two provided insight into what impact adding training data to specific classifications would have on the overall accuracy of classifications.

Commented [AG1]: S.-H. Park, M.-Y. Kim, Y.-J. Kim, and Y.-H. Park, "A Deep Learning Approach to Analyze Airline Customer Propensities: The Case of South Korea," *Applied Sciences*, vol. 12, no. 4, Art. no. 1916, 2022.

The third round added five additional examples to the Positive_Praise classification, thereby providing a total of 45 labeled examples in a balanced manner. These rounds were performed in sequence so that the study could assess the influence of the number of examples in the dataset on the model's ability to produce correct classifications.

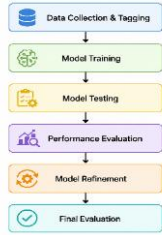


Fig. 1. Workflow of the proposed no-code deep learning text classification process

B. Model Training

The dataset that was tagged was uploaded into the Machine Learning for Kids platform to build an automated text classifier through a non-coding interface. As the platform trained on the examples of text with labels, it processed them automatically and made connections based on the relationship of word or phrase combinations to their respective label categories using a neural network. This is different from rule-based systems where you must write out manual if/else statements. The model was able to learn directly from the example data. At this time in the process, internal feature extraction was completed as textual input was converted to machine-readable form prior to the learning of the classification patterns.

C. Experimental Design

Three experiments were conducted to evaluate how the quantity and distribution of training data influenced model performance. Each experiment used the same three classification labels but differed in the number of training examples assigned to each category. Table I summarizes the training dataset used for each experimental iteration.

Experiment	Positive Praise	Negative Complaint	Urgent Support	Total
1	10	10	10	30
2	10	15	15	40
3	15	15	15	45

Table I. Training Dataset Configuration Across Experimental Iterations

D. Performance Evaluation

The trained model was tested with fifteen previously unseen airline customer comments that had been removed from the training set. Each comment was placed in one of three categories for testing: a sentence that has an obvious pattern; a statement that is either ambiguous or sarcastic; or a query that is an "out-of-vocabulary" inquiry about airlines. For each iteration of experimentation, the predicted category, confidence level, and whether correct classification occurred were documented. To compare results from the three different experiments, model performance was then analyzed based on the number and variety of examples included within

the labeled training data as it related to both the accuracy of classification and confidence levels.

III. RESULT AND DISCUSSION

Classification model performance evaluation. Three experiments were conducted to determine the effect that changes in training dataset size and distribution have on classifying airline customer feedback test cases. All three experiments utilized the same fifteen unseen airline customer feedback test cases; therefore, a consistent comparison of prediction accuracy and confidence was made across different training configurations. In addition to clear pattern inputs, the testing scenarios also included ambiguous or sarcastic, out-of-vocabulary (OOV) inputs. For every test case, the predicted category, confidence score, and classification correctness were recorded.

A. Experiment 1: Baseline Model (10-10-10)

The first experiment served as the baseline model using 10 tagged training examples for each of the three categories, resulting in a total of 30 labeled sentences. Table II summarizes the classification performance of the baseline model.

	TOTAL TEST CASES	CORRECT TEST	ACCURACY
Positive Praise	3	3	100.00%
Negative Complaint	6	5	83.33%
Urgent Support	3	3	100.00%
General Inquiry (Out-of-Vocabulary)	3	N/A	N/A

Table II. Classification Performance of Experiment 1

The baseline model had corrected predictions for both the Positive Praise and Urgent Support test cases. One sarcastic statement, "Great, another delayed flight. Just what I needed," was incorrectly predicted as Positive Praise because it recognized the term "Great" in the sentence as having a positive sentiment. The model correctly assigned all 3 out-of-vocabulary airline inquiries as being classified as Urgent Support, which indicates that the model is attempting to apply an unseen input (the unfamiliar terms) to its closest learned category rather than acknowledging them as new requests.

B. Experiment 2: Expanded Negative and Urgent Dataset (10-15-15)

To improve the recognition of challenging customer feedback, additional training examples were added to the Negative Complaint and Urgent Support categories, increasing each to 15 examples, while Positive Praise remained at 10 examples. Table III presents the performance of the refined model.

	TOTAL TEST CASES	CORRECT TEST	ACCURACY
Positive Praise	3	2	66.67%
Negative Complaint	6	6	100.00%
Urgent Support	3	3	100.00%
General Inquiry (Out-of-Vocabulary)	3	N/A	N/A

Table III. Classification Performance of Experiment 2

In addition to correcting the sarcastic example that was originally misclassified (and thus increasing the model's confidence in classifying that example as a negative complaint), the addition of these two new complaints increased the number of examples in the model. As a result, the model began to classify a real positive comment ("The flight exceeded my expectations, and the crew was amazing") as a "Negative Complaint." This decrease in positive praise classification accuracy from 100% to 66.67% illustrates that an improvement

in one category can create bias if all categories are not balanced.

C. Experiment 3: Balanced Dataset (15–15–15)

In the final experiment, the Positive_Praise category was also expanded to 15 training examples, resulting in a balanced dataset containing 45 labeled sentences. This configuration was intended to determine whether balancing the number of training examples across all categories would improve the model's overall performance.

	TOTAL TEST CASES	CORRECT TEST	ACCURACY
Positive Praise	3	2	66.67%
Negative Complaint	6	6	100.00%
Urgent Support	3	3	100.00%
General Inquiry (Out-of-Vocabulary)	3	N/A	N/A

Table IV. Classification Performance of Experiment 3

Balancing the data set allowed the model to correctly classify authentic positive feedback with accuracy in identifying sarcastic complaints as well as urgent support requests. In addition, the confidence level of predictions for each test case improved uniformly, showing that the model benefited from being trained on a more complete data set. While still assigning out-of-vocabulary queries to their nearest corresponding learned categories, the balanced data set generated the best overall classification results.

D. Explicit Rules vs. Tagged Learning

An important drawback of traditional methods of text classification is the use of "if-then" type rules to create all possible combinations of complaints made by airline customers. Many times, the same customer will write about the same concern (i.e., delayed flights) using completely different terminology; however, they are referring to the same problem. For example, when referring to a delayed flight, a customer might say: "My flight was late," "I had to wait hours until I boarded my flight," or "Great, just what I need, another delayed flight." It can be seen in this example how three completely different sentences refer to the same concern. Manually developing an algorithm to recognize all variations of sentences that describe a particular concern would be impossible and could never be maintained since there are continually new ways of expressing concerns.

Unlike the method, where we relied on the manual development of algorithms for each possible combination of complaints, the above model uses machine learning to train on labeled data. This allows the model to identify linguistic characteristics that distinguish between categories. After it identifies them, it applies to test cases that have been unlabeled. The results show that the model has correctly classified many different types of sentences. However, the misclassification of sarcasm from the first experiment indicates that the model's ability to classify depends on whether there are enough representative examples of sarcasm available during the training process.

E. Data Bias and Dataset Quality

The results from all experiments illustrate that the quality, quantity, and distribution of the training dataset have a considerable effect on how well a model can classify.

In Experiment 1, ten example sets for each category were used to train the model. The model classified all clearly defined customer statements with high levels of accuracy; however, it incorrectly classified a sarcastic complaint. A major factor in this classification error was the lack of sufficient representation within the training set for sarcasm.

In Experiment 2, an increase in training examples was added to only two categories (Negative_Complaint and Urgent_Support). As a result, there was a significant improvement in the recognition of sarcastic complaints by the model. There was also a great deal of confidence in the classifications made. An important finding in Experiment 2 was that although the addition of training examples for only two categories greatly enhanced the model's ability to recognize sarcastic complaints, it also caused the model to misclassify a true positive statement into the Negative_Complaint category. It appears that adding examples only to certain categories causes bias towards those categories. Therefore, enhancing one category at the expense of creating an imbalance within the entire dataset can have detrimental effects on the other categories.

Experiment 3 created a completely balanced dataset that contained fifteen example sets for every category. The model was again able to accurately classify positive customer feedback. Additionally, the model maintained its enhancements for negative complaints and urgent support requests. Overall, these results clearly show that deep learning models are highly dependent on adequate amounts of appropriately labeled and diverse training datasets.

F. Feature Extraction in the Deep Learning Pipeline

Feature extraction took place in the model training stage, as depicted in Figure 1. Once a user entered tagged airline customer feedback through the Machine Learning for Kids interface, the neural network immediately generated numerical representations from this textual input. The process of converting textual data into machine-readable numerical representations allowed the model to learn and identify common language usage (i.e., features) without having users pre-select keywords for inclusion or write specific code to define these aspects.

In terms of the above-mentioned experimental analysis, feature extraction provided an opportunity for the model to compare various customer feedback statements and find similarities across them based upon differences in word choice. However, the results of the analyses indicated that the characteristics of the features produced by the models were highly dependent on the datasets used for training purposes. For example, in the first experiment, since no sarcastic comments were included within the dataset, limited features were developed by the model to assist it in correctly identifying sarcasm in complaints. In contrast, once a diverse set of sarcastic comments was added to subsequent datasets in later experiments, the model's ability to develop a larger set of comprehensive features improved. Therefore, as demonstrated in both instances, both feature extraction and learning are directly impacted by the overall quality and diversity of the labeled training data.

IV. CONCLUSION

The primary purpose of this laboratory was to demonstrate how you can apply a no-code deep learning method to categorize positive praise, negative complaints, or urgent support issues with airline customer comments. Through iterative experimentation (three), this research has shown that a neural network learns based on examples that are previously identified as such rather than through manual IF/THEN programming, and that the quality, quantity, and distribution of these example inputs significantly impact the accuracy of classifications produced by the model. In addition to showing that models trained with a diverse number of examples will be able to recognize ambiguity within customer comments, the experiments also showed

that when training is done using balanced sets of examples, the potential for classification bias diminishes and the consistency of output improves. Overall, the lab work demonstrates the importance of high-quality training data, representatively diverse data sets, and continuous improvements to your model to develop accurate deep-learning-based text classification systems.

REFERENCES

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